

Gas Detection System Design & Sensor Placement

Complete Hazard Detection Design and Code Compliance for Industrial, Laboratory, and Specialty Gas Environments.

Our Expertise

Safe Gas Systems provides expert gas detection design and hazard mapping, backed by code knowledge, airflow analysis, and proven sensor strategies. Each solution is built around your site's hazards, ventilation, and regulations.

Our team designs for:

- ✓ Oxygen Deficiency Hazards (ODH)
- ✓ Flammable Gases (Hydrogen, Methane, LPG)
- ✓ Toxic Gases (CO, Cl₂, NH₃, H₂S)
- ✓ Asphyxiants and Elevated CO₂/N₂ Environments

What's Included

- ✓ Hazard and leak source mapping
- ✓ Airflow evaluation (smoke testing and CFD modeling)
- ✓ Sensor placement plans with mounting elevations
- ✓ Validation testing and commissioning support
- ✓ Alarm setpoint design and control logic support
- ✓ Coordination with the fire marshal and AHJ
- ✓ Deliverables: maps, logs, and traceable documentation



Codes & Standards

We design to current codes, standards, and industry best practices, including:

- ✓ OSHA 29 CFR 1910
- ✓ NFPA 72, IFC, and CGA Guidelines (TR-7, TR-8)
- ✓ IEC 60079-29-2 / ISO 14644-3 (airflow testing)
- ✓ ANSI/ASSP Z390 series and confined space protocols

Applications by Industry

- ✓ Cryogenic Storage & Distribution Rooms
- ✓ CO₂ Beverage Systems / Food & Beverage Plants
- ✓ Hydrogen & CNG Repair Garages / Test Facilities
- ✓ Research Labs / Clean Rooms
- ✓ Medical Gas & Hospital Backfeed Areas

Why Choose SGS?

- ✓ Industry-recognized gas detection design expertise
- ✓ Proven CFD airflow modeling capabilities
- ✓ AHJ-ready documentation and validation
- ✓ Works with all major detection brands (Honeywell, Dräger, MSA, and more)

Contact Us

Safe Gas Systems

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SGS designs gas detection systems that protect your people, equipment, and compliance record.

